

Operators and Control Statements & Looping

INSERTION SORT

The screenshot displays the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows a list of questions, with 'Sort-I' selected. The main area is divided into three sections: 'PROGRAM EDITOR', 'OUTPUT', and 'INPUT'.

PROGRAM EDITOR: Contains the following Java code for Insertion Sort:

```
1 import java.util.Scanner;
2 public class insertion{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int arr[]=new int[n];
7         System.out.println("enter the elements:");
8         for(int i=0;i<n;i++){
9             arr[i]=sc.nextInt();
10        }
11        for(int i=1;i<n;i++){
12            int temp=arr[i];
13            int j;
14            for(j=i-1;j>=0&&arr[j]>temp;j--){
15                arr[j+1]=arr[j];
16            }
17            arr[j+1]=temp;
18        }
19        System.out.println("enter sorted array:");
20        for(int i=0;i<n;i++){
21            System.out.print(" " + arr[i]);
22        }
23    }
24 }
```

OUTPUT: Shows the execution results:

```
enter the elements:
enter sorted array:
3 6 7 9 10
```

INPUT: Shows the input values entered by the user:

```
5
3
9
10
6
7
```

BUBBLE SORT

The screenshot displays the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The left sidebar shows a list of questions, with 'Sort-B' selected. The main area is divided into three sections: 'PROGRAM EDITOR', 'OUTPUT', and 'INPUT'.

PROGRAM EDITOR: Contains the following Java code for Bubble Sort:

```
1 import java.util.Scanner;
2 public class sort{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int a[]=new int[n];
7         System.out.println("Enter the five elements :");
8         for(int i=0;i<n;i++){
9             a[i]=sc.nextInt();
10        }
11        for(int i=0;i<n-1;i++){
12            for(int j=0;j<n-1-i;j++){
13                if(a[j]>a[j+1]){
14                    int temp=a[j];
15                    a[j]=a[j+1];
16                    a[j+1]=temp;
17                }
18            }
19        }
20        System.out.println("sorted array");
21    }
22 }
```

OUTPUT: Shows the execution results:

```
Enter the five elements :
sorted array
9 8 6 5 1
```

INPUT: Shows the input values entered by the user:

```
5
1
9
5
8
6
```

2ND MIN



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JAVA PROGRAMMING

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192425211RC

▼

QUESTIONS 31 / 31 completed

2nd Min ▼

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java ▼

Run Save

```
1 import java.util.Scanner;
2 public class secondmin{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println(" enter number elements");
6         int n=sc.nextInt();
7         int arr[]=new int[n];
8         System.out.print("enter the numbers in the array:");
9         for(int i=0;i<n;i++){
10             arr[i]=sc.nextInt();
11         }
12         int smallest=arr[0];
13         int second=arr[0];
14         for(int i=1;i<arr.length;i++){
15             if(arr[i]<smallest){
16                 second=smallest;
17                 smallest=arr[i];
18             }
19             else if(arr[i]<second&&arr[i]!=smallest){
20                 second=arr[i];
21             }
22         }
23     }
24 }
```

OUTPUT

enter number elements
enter the numbers in the array:enter the second smallest number:1

INPUT

5
1
2
3
4
0

MIN



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▼

QUESTIONS 31 / 31 completed

Min ▼

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java ▼

Run Save

```
1 import java.util.Scanner;
2 public class min{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter the number of element:");
6         int n=sc.nextInt();
7         int arr[]=new int[n];
8         for(int i=0;i<n;i++){
9             arr[i]=sc.nextInt();
10        }
11        int min=arr[0];
12        for(int i=1;i<n;i++){
13            if(arr[i]<min){
14                min=arr[i];
15            }
16        }
17        System.out.print("minimum number : " + min);
18    }
19 }
```

OUTPUT

enter the number of element:
minimum number :1

INPUT

5
1
2
3
4
5

2ND MAX

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QUESTIONS 31 / 31 completed

2nd Max

2ND MAX
Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class second_max{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter the number of elements:");
6         int n=sc.nextInt();
7         int arr[]=new int[n];
8         System.out.println("enter the elements in the array :");
9         for(int i=0;i<n;i++){
10             arr[i]=sc.nextInt();
11         }
12         int largestnumber=arr[0];
13         int secondlargestnumber=arr[0];
14         for(int i=1;i<n;i++){
15             if(arr[i]>largestnumber){
16                 secondlargestnumber=largestnumber;
17                 largestnumber=arr[i];
18             }
19             else if(arr[i]>secondlargestnumber&&arr[i]!=largestnumber){
20                 secondlargestnumber=arr[i];
21             }
22         }
23     }
24 }
```

INPUT

5
1
2
3
4
5

OUTPUT

enter the number of elements:
enter the elements in the array :
enter the second largest number:4

MAX

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QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY
Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class max{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n=sc.nextInt();
7         int arr[]=new int[n];
8         System.out.println("enter the elements :");
9         for(int i=0;i<n;i++){
10             arr[i]=sc.nextInt();
11         }
12         int max=arr[0];
13         for(int i=1;i<n;i++){
14             if(arr[i]>max){
15                 max=arr[i];
16             }
17         }
18         System.out.println("maximum number in the array is " + max);
19     }
20 }
```

INPUT

5
1
2
3
4
5

OUTPUT

enter the number of elements:enter the elements :
maximum number in the array is 1

PERFECT



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QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class first
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int sum=0;
9         for(int i=1;i<a;i++){
10             if(a%i==0){
11                 sum=sum+i;
12             }
13         }
14         if(sum==a){
15             System.out.println("PerFect number");
16         }else{
17             System.out.println("not perfect number");
18         }
19     }
20 }
```

OUTPUT

not perfect number

INPUT

28

RT -NUM PATTERN



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QUESTIONS 31 / 31 completed

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:
1
22
333
4444
55555

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class rt{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         for(int i=1;i<=n;i++){
7             for(int j=1;j<=i;j++){
8                 System.out.print(" " + i);
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

1
12
123
1234
12345

INPUT

5

SPHERE

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", the subtitle "JAVA PROGRAMMING", and a user profile for "KARNATI UDAY KIRAN REDDY" with ID "192425211RC". The left sidebar shows "QUESTIONS 31 / 31 completed" and a dropdown menu with "Sphere" selected. Below this, the problem description for "SHPERE" is displayed: "Write a Java program to calculate the volume of a sphere using the radius input by the user." The main area is the "PROGRAM EDITOR" with a language dropdown set to "Java". It contains a Java program to calculate the volume of a sphere. The "INPUT" field on the right contains the value "5". The "OUTPUT" field at the bottom shows the result: "Volume of sphere 392.69908169872417".

QUESTIONS 31 / 31 completed

Sphere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class first
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int radius=sc.nextInt();
8         double volume=(4/3)*3.14*radius*radius*radius;
9         System.out.println("Volume of sphere= " + volume);
10    }
11 }
```

INPUT

5

OUTPUT

Volume of sphere 392.69908169872417

ARMSTRONG

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", the subtitle "JAVA PROGRAMMING", and a user profile for "KARNATI UDAY KIRAN REDDY" with ID "192425211RC". The left sidebar shows "QUESTIONS 31 / 31 completed" and a dropdown menu with "Armstrong" selected. Below this, the problem description for "ARMSTRONG" is displayed: "Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements." The main area is the "PROGRAM EDITOR" with a language dropdown set to "Java". It contains a Java program to check if a number is an Armstrong number. The "INPUT" field on the right contains the value "153". The "OUTPUT" field at the bottom shows the result: "not armstrong number".

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class armstrong
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int n=sc.nextInt();
8         int original=n;
9         int sum=0;
10        int digit=0;
11        while(n!=0){
12            digit=n%10;
13            sum=sum+digit*digit*digit;
14            n=n/10;
15        }
16        if(sum==original)
17            System.out.println("armstrong number");
18        else
19            System.out.println("not armstrong number");
20    }
21 }
```

INPUT

153

OUTPUT

not armstrong number

PALINDROME

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name. The left sidebar shows the 'Palindrom' question selected. The main area contains a Java program editor with the following code:

```
1 import java.util.Scanner;
2 public class first{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int rev=0;
7         int original=a;
8         while(a>0){
9             rev=rev*10+a%10;
10            a=a/10;
11        }
12        if(original==rev){
13            System.out.println("Palindrome");
14        }else{
15            System.out.println("not a Palindrome");
16        }
17    }
18 }
```

The input field contains '123321' and the output field displays 'not a Palindrome'.

POWER

The screenshot shows the SIMATS Online Programming Lab interface. The left sidebar shows the 'Power' question selected. The main area contains a Java program editor with the following code:

```
1 import java.util.Scanner;
2 public class power{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int base=sc.nextInt();
6         int exponent=sc.nextInt();
7         int result=1;
8         for(int i=1;i<=exponent;i++){
9             result=result*base;
10        }
11        System.out.println("result " + result);
12        sc.close();
13    }
14 }
```

The input field contains '2' and '6', and the output field displays 'result 64'.

PRODUCT



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QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class product{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int product=1;
7         while(n>0){
8             product=product*(n%10);
9             n=n/10;
10        }
11
12        System.out.println("product " + product);
13    }
14 }
```

OUTPUT

product 24

INPUT

234

SUM



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QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class sum{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int sum=0;
7         while(a>0){
8             int digit=a%10;
9             sum=sum+digit;
10            a=a/10;
11        }
12        System.out.println("Sumof the digits " + sum);
13    }
14 }
```

OUTPUT

Sumof the digits 5

INPUT

5

B TO D

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user profile "KARNATI UDAY KIRAN REDDY". The main section is titled "PROGRAM EDITOR" and contains a Java program for converting a binary number to a decimal number. The program uses a while loop to calculate the decimal value by multiplying each binary digit by its corresponding power of 10 and then summing the results. The input field on the right contains the binary number "1001". The output field shows the result "decimal number is : 9".

QUESTIONS 31 / 31 completed

B to D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class decimal{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter the binary number:");
6         int binary=sc.nextInt();
7         int decimal=0;
8         int power=1;
9         while(binary>0){
10             int digit=binary%10;
11             decimal=decimal+digit*power;
12             power=power*2;
13             binary=binary/10;
14         }
15         System.out.println("decimal number is :"+ decimal);
16     }
17 }
```

INPUT

1001

OUTPUT

enter the binary number:
decimal number is : 9

D TO B

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user profile "KARNATI UDAY KIRAN REDDY". The main section is titled "PROGRAM EDITOR" and contains a Java program for converting a decimal number to a binary number. The program uses a while loop to calculate the binary value by dividing the decimal number by 2 and storing the remainder. The input field on the right contains the decimal number "10". The output field shows the result "Binary Number : 1010".

QUESTIONS 31 / 31 completed

D to B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class binary{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter the number:");
6         int n=sc.nextInt();
7         int binary=0;
8         int place=1;
9         while(n>0){
10             int remainder=n%2;
11             binary=binary+remainder*place;
12             place=place*10;
13             n=n/2;
14         }
15         System.out.println("Binary Number :"+ binary);
16     }
17 }
```

INPUT

10

OUTPUT

enter the number:
Binary Number : 1010

LCM

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and a user profile for KARNATI UDAY KIRAN REDDY. The left sidebar shows "QUESTIONS 31/31 completed" and a dropdown menu with "LCM" selected. Below this, the problem description for LCM is provided. The main area is divided into three sections: "PROGRAM EDITOR" with a Java code editor, "INPUT" with a text area containing "10" and "20", and "OUTPUT" with a text area containing "Lcm 20".

QUESTIONS 31/31 completed

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class lcm{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int b=sc.nextInt();
7         int gcd=1;
8         for(int i=1;i<=a&&i<=b;i++){
9             if(a%i==0&&b%i==0){
10                 {
11                     gcd=i;
12                 }
13             }
14         }
15         int lcm=(a*b)/gcd;
16         System.out.println("Lcm " + lcm);
17     }
```

INPUT

10
20

OUTPUT

Lcm 20

LT -PATTERN

The screenshot shows the SIMATS Online Programming Lab interface for the LT-PATTERN problem. The top header is the same as the previous screenshot. The left sidebar shows "QUESTIONS 31/31 completed" and a dropdown menu with "LT-Pattern" selected. Below this, the problem description for LT-PATTERN is provided. The main area is divided into three sections: "PROGRAM EDITOR" with a Java code editor, "INPUT" with a text area containing "5", and "OUTPUT" with a text area showing a pattern of asterisks.

QUESTIONS 31/31 completed

LT-Pattern

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class pattern{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         for(int i=n;i>=1;i--){
7             for(int j=1;j<=i;j++){
8                 System.out.print("** ");
9             }
10            System.out.println();
11        }
12    }
13 }
```

INPUT

5

OUTPUT

```
*****
****
***
**
*
```

FACTORIAL



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JAVA PROGRAMMING

QUESTIONS 31/31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

```
1 import java.util.Scanner;
2 public class recursion
3 {
4     static int factorial(int n){
5         if(n==0||n==1)
6             return 1;
7         else
8             return n*factorial(n-1);
9     }
10    public static void main(String[] args)
11    {
12        Scanner sc=new Scanner(System.in);
13        int num=sc.nextInt();
14        if(num<0){
15            System.out.println("Factorial is not defined ");
16        }
17        else{
18            System.out.println("Factorial of" + num + "=" + factorial(num));
19        }
20    }
21 }
```

OUTPUT

Factorial 120

INPUT

5

TABLE



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JAVA PROGRAMMING

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

```
1 import java.util.Scanner;
2 public class first
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         for(int i=1;i<=10;i++){
9             System.out.println(a + "X" + i + "=" + a*i);
10        }
11    }
12 }
```

OUTPUT

5X1=5
5X2=10
5X3=15
5X4=20
5X5=25
5X6=30

INPUT

5

SELECTION SORT

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "KARNATI UDAY KIRAN REDDY" with ID "192425211RC". The "JAVA PROGRAMMING" tab is selected. On the left, the "Sort" category is chosen, displaying the "SORT" problem description: "Write a Java program to implement selection sort on an array of integers using loops and conditional statements." The main editor shows a Java program for selection sort. The code includes imports, class definition, and a main method that reads an array of integers and sorts it using selection sort logic. The output section shows the program's execution with input values 5, 10, 78, 9, 3, 1 and the resulting sorted array 1, 3, 9, 10, 78.

```
1 import java.util.Scanner;
2 public class selection{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int arr[]=new int[n];
7         System.out.println("Enter the elements :");
8         for(int i=0;i<n;i++){
9             arr[i]=sc.nextInt();
10        }
11        for(int i=0;i<n-1;i++){
12            for(int j=0;j<n-1-i;j++){
13                if(arr[j]<arr[j+1]){
14                    int temp=arr[j];
15                    arr[j]=arr[j+1];
16                    arr[j+1]=temp;
17                }
18            }
19        }
20        System.out.println("Enter the sorted array:");
21        for(int i=0;i<n;i++){
```

OUTPUT

```
Enter the elements :
Enter the sorted array:
78
10
3
9
1
```

PRIME

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "KARNATI UDAY KIRAN REDDY" with ID "192425211RC". The "JAVA PROGRAMMING" tab is selected. On the left, the "Prime" category is chosen, displaying the "PRIME" problem description: "Write a Java program to check whether a given number is a prime number or not using loops and conditional statements." The main editor shows a Java program for checking prime numbers. The code includes imports, class definition, and a main method that reads an integer and checks if it is prime by testing divisibility from 2 to the number itself. The output section shows the program's execution with input value 7 and the resulting output "non Prime Number".

```
1 import java.util.Scanner;
2 public class prime
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int count=0;
9         for(int i=1;i<=a;i++)
10        {
11            if(a%i==0){
12                count ++;
13            }
14        }
15        if(count==2)
16            System.out.println("Prime number");
17        else
18            System.out.println("non Prime Number");
19    }
20 }
```

OUTPUT

```
non Prime Number
```

GCD

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and a user profile for KARNATI UDAY KIRAN REDDY. The left sidebar shows "QUESTIONS 31/31 completed" and a dropdown menu with "GCD" selected. Below this, the problem description states: "Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements." The main area is the "PROGRAM EDITOR" with a Java code editor containing the following code:

```
1 import java.util.Scanner;
2 public class gcd{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a=sc.nextInt();
6         int b=sc.nextInt();
7         int gcd=1;
8         for(int i=1;i<=a&&i<=b;i++){
9             if(a%i==0&&b%i==0){
10                 gcd=i;
11             }
12         }
13         System.out.println("Gcd " + gcd);
14     }
15 }
```

Below the editor is the "OUTPUT" section showing "Gcd: 30". On the right, the "INPUT" section shows the numbers "30" and "60".

BINARY SEARCH

The screenshot shows the SIMATS Online Programming Lab interface for the BINARY SEARCH problem. The header is identical to the GCD problem. The left sidebar shows "QUESTIONS 31/31 completed" and a dropdown menu with "Search" selected. Below this, the problem description states: "Write a Java program to implement binary search on an array of integers using loops and conditional statements." The main area is the "PROGRAM EDITOR" with a Java code editor containing the following code:

```
1 import java.util.Scanner;
2 public class search
3 {
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         System.out.print("enter number of elements:");
7         int n=sc.nextInt();
8         int arr[]=new int[n];
9         System.out.println("enter sorted array elements : ");
10        for(int i=0;i<n;i++){
11            arr[i]=sc.nextInt();
12        }
13        System.out.print("enter element to search :");
14        int key=sc.nextInt();
15        int low=0;
16        int high=n-1;
17        int position=-1;
18        while(low<=high){
19            int mid=(low+high)/2;
20            if(arr[mid]==key){
21                position=mid;
22            }
23        }
24    }
25 }
```

Below the editor is the "OUTPUT" section showing the following text:

```
enter number of elements:enter sorted array elements :
enter element to search :element found at index:2
```

On the right, the "INPUT" section shows a list of numbers: "5", "10", "20", "30", "40", "50", "30".

RT-PATTERN

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and the user "KARNATI UDAY KIRAN REDDY" with ID "1924252118C".

On the left, a sidebar shows a dropdown menu with "RT-Pattern" selected. Below it, the "RT-PATTERN" section contains the instruction: "Write a Java program to print the following pattern using loops:" followed by a pattern of asterisks:

```
**  
***  
****  
*****
```

The main area features a "PROGRAM EDITOR" with the following Java code:

```
1 import java.util.Scanner;  
2 public class first  
3 {  
4     public static void main(String[] args)  
5     {  
6         Scanner sc=new Scanner(System.in);  
7         int a=sc.nextInt();  
8         for(int i=1;i<=5;i++)  
9         {  
10            for(int j=1;j<=i;j++)  
11            {  
12                System.out.print("** ");  
13            }  
14            System.out.println();  
15        }  
16        sc.close();  
17    }  
18 }
```

Below the editor is an "OUTPUT" section showing the pattern of asterisks. To the right of the editor is an "INPUT" field containing the number "5".

FACTORIAL

The screenshot shows the SIMATS Online Programming Lab interface. The header is identical to the previous one.

On the left, a sidebar shows a dropdown menu with "Factorial" selected. Below it, the "FACTORIAL" section contains the instruction: "Write a Java program to calculate the factorial of a given number using loops."

The main area features a "PROGRAM EDITOR" with the following Java code:

```
1 import java.util.Scanner;  
2 public class main  
3 {  
4     public static void main(String[] args)  
5     {  
6         Scanner sc=new Scanner(System.in);  
7         int num=sc.nextInt();  
8         int fact=1;  
9         for(int i=1;i<=num;i++)  
10        {  
11            fact=fact*i;  
12        }  
13        System.out.println("factorial " + fact);  
14    }  
15 }
```

Below the editor is an "OUTPUT" section showing the result: "factorial120". To the right of the editor is an "INPUT" field containing the number "5".



SIMATS Online Programming Lab

JAVA PROGRAMMING



KARNATI UDAY KIRAN REDDY
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QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class first
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int b=sc.nextInt();
9         int c=a+b;
10        int d=a-b;
11        int e=a*b;
12        int f=a/b;
13        System.out.println("sum = " + c);
14        System.out.println("Difference = " + d);
15        System.out.println("product = " + e);
16        System.out.println("division = " + f);
17    }
18 }
```

OUTPUT

```
sum = 14
```

INPUT

5
9